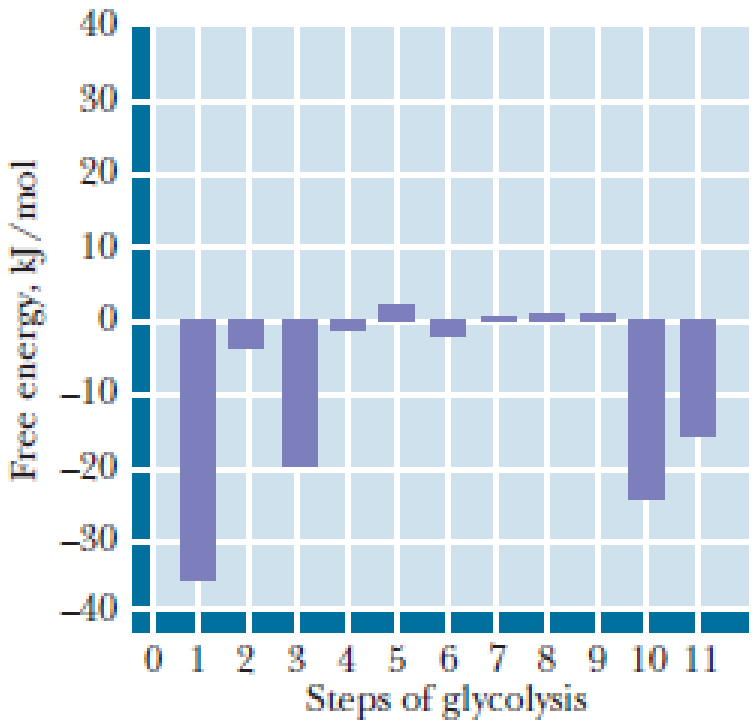
Summary of Glycolysis

Reaction	Enzyme	$\Delta G^{\circ\prime}$ (kJ/mol)	K_{eq} at 25°C	ΔG (kJ/mol)
α-D-Glucose + ATP^{4-} ⇌ glucose-6-phosphate^{2-} + ADP^{3-} + H^+	Hexokinase Glucokinase	−16.7	850	−33.9*
Glucose-6-phosphate^{2-} ⇌ fructose-6-phosphate^{2-}	Phosphoglucoisomerase	+1.67	0.51	−2.92
Fructose-6-phosphate^{2-} + ATP^{4-} ⇌ fructose-1,6-bisphosphate^{4-} + ADP^{3-} + H^+	Phosphofructokinase	−14.2	310	−18.8
Fructose-1,6-bisphosphate^{4-} ⇌ dihydroxyacetone-P^{2-} + glyceraldehyde-3-P^{2-}	Fructose bisphosphate aldolase	+23.9	6.43×10^{-5}	−0.23
Dihydroxyacetone-P^{2-} ⇌ glyceraldehyde-3-P^{2-}	Triose phosphate isomerase	+7.56	0.0472	+2.41
Glyceraldehyde-3-P^{2-} + P_i^{2-} + NAD^+ ⇌ 1,3-bisphosphoglycerate^{4-} + NADH + H^+	Glyceraldehyde-3-P dehydrogenase	+6.30	0.0786	−1.29
1,3-Bisphosphoglycerate^{4-} + ADP^{3-} ⇌ 3-P-glycerate^{3-} + ATP^{4-}	Phosphoglycerate kinase	−18.9	2060	+0.1
3-Phosphoglycerate^{3-} ⇌ 2-phosphoglycerate^{3-}	Phosphoglycerate mutase	+4.4	0.169	+0.83
2-Phosphoglycerate^{3-} ⇌ phosphoenolpyruvate^{3-} + H_2O	Enolase	+1.8	0.483	+1.1
Phosphoenolpyruvate^{3-} + ADP^{3-} + H^+ ⇌ pyruvate$^-$ + ATP^{4-}	Pyruvate kinase	−31.7	3.63×10^5	−23.0
Pyruvate$^-$ + NADH + H^+ ⇌ lactate$^-$ + NAD^+	Lactate dehydrogenase	−25.2	2.63×10^4	−14.8



- For reactions 2 and 4 through 9, Δ*G is very close to zero, meaning these reactions operate essentially at equilibrium.*
- Small changes in the concentrations of reactants and products could "push" any of these reactions either forward or backward.
- By contrast, the hexokinase, phosphofructokinase, and pyruvate kinase reactions all exhibit large negative *DG values under* cellular conditions.
- These reactions are thus the sites of glycolytic regulation. When these three enzymes are active, glycolysis proceeds and glucose is readily metabolized to pyruvate or lactate.

Summary – balance sheet

The ATP and NADH balance sheet for glycolysis

Number[a]	Reaction per Glucose	ATP Change per Glucose[b]	NADH Change per Glucose[b]
1	Glucose → glucose-6-phosphate	−1	0
3	Fructose-6-phosphate → fructose-1,6-bisphosphate	−1	0
6	2 Glyceraldehyde-3-phosphate ⇌ 2 1,3-bisphosphoglycerate	0	+2
7	2 1,3-Bisphosphoglycerate ⇌ 2 3-phosphoglycerate	+2	0
10	2 Phosphoenolpyruvate → 2 pyruvate	+2	0
Total change		+2	+2

[a] The number corresponds to the reaction number in Table 15.1.

[b] A minus sign indicates loss of ATP by cleavage of a phosphoanhydride bond; a plus sign indicates formation of ATP (from ADP) or NADH (from NAD^+).

Summary of citric acid cycle

Reactions of the TCA Cycle

Reaction	Enzyme	$\Delta G^{\circ\prime}$ (kcal/mol)
1. Acetyl-CoA + oxaloacetate + H_2O $\longrightarrow$ citrate + CoA	Citrate synthase	−7.7
2. Citrate $\rightleftharpoons$ *cis*-aconitase $\rightleftharpoons$ isocitrate	Aconitase	+1.5
3. Isocitrate + NAD^+ $\longrightarrow$ α-ketoglutarate + NADH + CO_2	Isocitrate dehydrogenase	−5.0
4. α-Ketoglutarate + NAD^+ + CoA $\longrightarrow$ Succinyl-CoA + NADH + CO_2	α-Ketoglutarate dehydrogenase	−8.0
5. Succinyl-CoA + GDP + P_i $\longrightarrow$ succinate + GTP + CoA	Succinate thiokinase	−0.7
6. Succinate + FAD $\longrightarrow$ fumarate + $FADH_2$	Succinate dehydrogenase	0.0
7. Fumarate + H_2O $\longrightarrow$ L-malate	Fumarase	−0.9
8. L-malate + NAD^+ $\longrightarrow$ oxaloacetate + NADH + H^+	Malate dehydrogenase	+7.1
Acetyl-CoA + 2 H_2O + 3 NAD^+ + FAD + ADP + P_i $\longrightarrow$ 2 CO_2 + 3 NADH + 2 H^+ + $FADH_2$ + CoA + ATP		−13.7

2* (3 NADH + $FADH_2$ + 1 ATP)

How many ATP's are produced from Kreb's cycle (one cycle)

- The number of ATP molecules produced by the electron transport system from Krebs cycle intermediates for once cycle can be calculated.

- A cycle for Krebs cycle produces 3 molecules of reduced coenzyme NADH2 and 1 molecule of reduced FADH2 besides 1 molecule of GTP is synthesized directly through substrate level phosphorylation (3:1:1).

- The reduced coenzymes (NADH2 and FADH2) are oxidised by mitochondrial electron transport system in the inner mitochondrial membrane.

✓ *A molecule of NADH2 when undergoes oxidation produces 3 molecules of ATP*
✓ *A molecule of FADH2 when undergoes oxidation produces 2 molecules of ATP*

- Thus the total ATP produced on oxidation of reduced coenzymes produced in one turn of Krebs cycle are (3x3) + (2x1) + 1 i.e., 12 ATP molecules.

Final balance sheet– No of ATP's from 1 mole of glucose

Balance Sheet for Oxidation of One Molecule of Glucose

	NADH	$FADH_2$	ATP
CYTOPLASMIC REACTIONS			
Glycolysis			
1. Glucose ⟶ glucose-6-P			−1
2. Fructose-6-P ⟶ fructose-1,6-P			−1
3. (2) Glyceraldehyde-3-P ⟶ glycerate-1,3-bisphosphate	+2		
4. (2) Glycerate-1,3-bisphosphate ⟶ glycerate-3-P			+2
5. (2) Phosphoenolpyruvate ⟶ pyruvate			+2
MITOCHONDRIAL REACTIONS			
Pyruvate Dehydrogenase			
(2) Pyruvate ⟶ acetyl-CoA	+2		
TCA Cycle			
1. (2) Isocitrate ⟶ α-ketoglutarate	+2		
2. (2) α-Ketoglutarate ⟶ succinyl-CoA	+2		
3. (2) Succinyl-CoA ⟶ succinate			+1.5[a]
4. (2) Succinate ⟶ fumarate		+2	
5. (2) Malate ⟶ oxaloacetate	+2		
Electron Transport–Oxidative Phosphorylation			
1. Reoxidation of NADH produced in glycolysis	−2		+3[b]
2. Reoxidation of NADH produced by pyruvate dehydrogenase	−2		+5
3. Reoxidation of NADH produced in TCA cycle	−6		+15
4. Reoxidation of $FADH_2$ produced in TCA cycle		−2	+3
Totals	0	0	29.5

[a] This number is 1.5 instead of 2 to take into account the energy required to export ATP to the cytosol.

[b] This number is 3 instead of 5 because the glycerol phosphate shuttle bypasses complex I.

Calculation of number of ATPs.

1. Glycolysis = 2 ATP + 2 NADH
   = 2 ATP + 6 ATP = 8 ATPs

Note:

1 NADH = 3 ATP.
1 FADH2 = 2 ATP

2. TCA + ETC cycle
   = 6 NADH + 2 FADH2 + 2 ATP
   = 6(3 ATP) + 2 (2 ATP) + 2 ATP
   = 18 + 4 + 2 ATPs
   = 26 ATPs

Total ATP is 8 + 26 = 34 ATPs

Why ATP number varies?

The number of ATP molecules generated from the catabolism of glucose varies. The reasons are

1. The number of hydrogen ions that the electron transport chain complexes can pump through the membrane varies between species.

2. Another source of variance stems from the shuttle of electrons across the mitochondrial membrane. The NADH generated from glycolysis cannot easily enter mitochondria. Thus, electrons are picked up on the inside of the mitochondria by either NAD^+ or FAD^+.

3. Fewer ATP molecules are generated when FAD^+ acts as a carrier. NAD^+ is used as the electron transporter in the liver and FAD^+ in the brain, so ATP yield depends on the tissue being considered.

4. Another factor that affects the yield of ATP molecules generated from glucose is that intermediate compounds in these pathways are used for other purposes. Glucose catabolism connects with the pathways that build or break down all other biochemical compounds in cells, and the result is somewhat messier than the ideal situations described thus far.

5. For example, sugars other than glucose are fed into the glycolytic pathway for energy extraction. Other molecules that would otherwise be used to harvest energy in glycolysis or the citric acid cycle may be removed to form nucleic acids, amino acids, lipids, or other compounds. Overall, in living systems, these pathways of glucose catabolism extract about 34 percent of the energy contained in glucose.

Oxidation – Reduction

Oxidation-Reduction Reactions The NADH dehydrogenase complex of the mitochondrial respiratory chain promotes the following series of oxidation-reduction reactions, in which Fe^{3+} and Fe^{2+} represent the iron in iron-sulfur centers, Q is ubiquinone, QH_2 is ubiquinol, and E is the enzyme:

(1) $NADH + H^+ + E\text{-}FMN \longrightarrow NAD^+ + E\text{-}FMNH_2$

(2) $E\text{-}FMNH_2 + 2Fe^{3+} \longrightarrow E\text{-}FMN + 2Fe^{2+} + 2H^+$

(3) $2Fe^{2+} + 2H^+ + Q \longrightarrow 2Fe^{3+} + QH_2$

Sum: $NADH + H^+ + Q \longrightarrow NAD^+ + QH_2$

For each of the three reactions catalyzed by the NADH dehydrogenase complex, identify **(a)** the electron donor, **(b)** the electron acceptor, **(c)** the conjugate redox pair, **(d)** the reducing agent, and **(e)** the oxidizing agent.

Oxidation – Reduction

Answer Oxidation-reduction reactions require an electron donor and an electron acceptor. Recall that electron donors are reducing agents; electron acceptors are oxidizing agents.

(1) NADH is the electron donor **(a)** and the reducing agent **(d);** E-FMN is the electron acceptor **(b)** and the oxidizing agent **(e);** NAD^+/NADH and E-FMN/E-$FMNH_2$ are conjugate redox pairs **(c).**

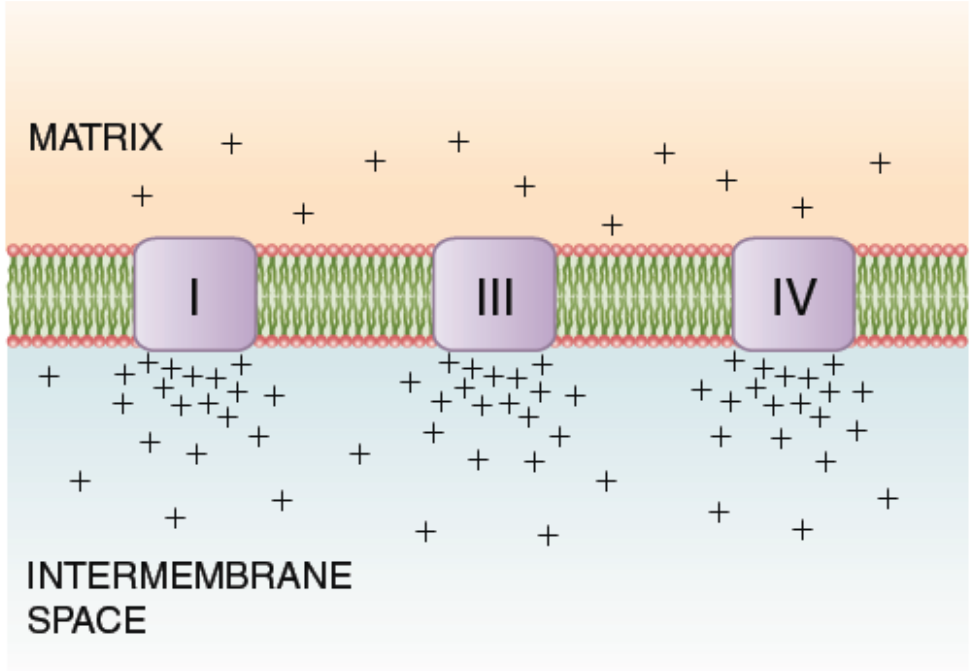
(2) E-$FMNH_2$ is the electron donor **(a)** and reducing agent **(d);** Fe^{3+} is the electron acceptor **(b)** and oxidizing agent **(e);** E-FMN/E-$FMNH_2$ and Fe^{3+}/Fe^{2+} are redox pairs **(c).**

(3) Fe^{2+} is the electron donor **(a)** and reducing agent **(d);** Q is the electron acceptor **(b)** and oxidizing agent **(e);** and Fe^{3+}/Fe^{2+} and Q/QH_2 are redox pairs **(c).**

Proton motive force (PMF)

- **ATP synthase molecules are the only place where H^+ can diffuse back to the matrix**.
  - ✓ The exergonic flow of H^+ is used by the enzyme to generate ATP. This coupling of the redox reactions of the electron transport chain to ATP synthesis is an example of chemiosmosis.
- How does the electron transport chain pump protons?
  - ✓ Certain members of the electron transport chain accept and release H^+ along with electrons.
  - ✓ At certain steps along the chain, electron transfers cause H^+ to be taken up and released into the surrounding solution.
  - ✓ The electron carriers are spatially arranged in the membrane in such a way that protons are accepted from the mitochondrial matrix and deposited in the intermembrane space.
- The H^+ gradient that results is the **proton-motive force**, a gradient with the capacity to do work.
  - The force drives H^+ back across the membrane through the specific H^+ channels provided by ATP synthases.

Proton motive force



- When the mitochondrial complexes translocate protons across the inner mitochondrial membrane, the concentration of H^+ outside increases and the concentration of H^+ inside decreases.

- *This imbalance of protons, a nonequilibrium state, has an associated free energy (the force that would restore the system to equilibrium).*

- The free energy change for generating the chemical imbalance of protons is

$$\Delta G = RT \ln \frac{[H^+]_{out}}{[H^+]_{in}}$$

- The free energy change for generating the electrical imbalance of protons is

$$\Delta G = Z\mathcal{F}\Delta\psi$$

*where **Z** is the ion's charge (+1 in this case) and Δψ is the membrane potential caused by the imbalance in positive). For mitochondria, Δψ is positive, usually 150 to 200 mV. This value indicates that the intermembrane space or cytosol is more positive than the matrix.*

- Combining the chemical and electrical effects gives an overall free energy change for transporting protons from the matrix (*in*) to the intermembrane space (*out*):

$$\Delta G = RT \ln \frac{[H^+]_{out}}{[H^+]_{in}} + Z\mathcal{F}\Delta\psi$$

Proton motive force

- Typically, the free energy change for translocating one proton out of the matrix is about +20 kJ. mol^{-1} .

Problem: Calculate the free energy change for translocating a proton out of the mitochondrial matrix, where pH_{matrix} = 7.8, $pH_{cytosol}$ = 7.15, $\Delta\psi$ = 170 mV, and T = 25°C.

Proton motive force

- Typically, the free energy change for translocating one proton out of the matrix is about +20 kJ. mol^{-1} .

Problem: Calculate the free energy change for translocating a proton out of the mitochondrial matrix, where pH_{matrix} = 7.8, $pH_{cytosol}$ = 7.15, $\Delta\psi$ = 170 mV, and T = 25°C.

Since pH = –log [H+], the logarithmic term of PMF Equation then becomes

$\Delta G = 2.303\, RT\, (pH_{in} - pH_{out}) + ZF\Delta\psi$

Substituting known values gives

ΔG = 2.303(8.3145 J · K−1 · mol−1)(298 K)(7.8 − 7.15) + (1)(96,485 J · V−1 · mol−1)(0.170 V)

= 3700 J · mol−1 + 16,400 J · mol−1

= +20.1 kJ · mol−1